

Tutorial Session: K-Means Clustering and PCA

What is K-Means?

K-Means is an unsupervised machine learning algorithm used for clustering data points into groups.

The algorithm divides the dataset into K clusters such that:

- ▶ Data points inside the same cluster are similar.
- ▶ Data points in different clusters are dissimilar.
- ▶ Similarity is measured using Euclidean distance.

Intuition Behind K-Means

Suppose we have customer data and want to group customers with similar behavior.

K-Means works as follows:

1. Randomly select K centroids.
2. Assign each point to the nearest centroid.
3. Update centroid positions.
4. Repeat until cluster assignments stop changing.

Step 1: Choose K

Examples:

- ▶ $K = 2$ means 2 clusters.
- ▶ $K = 3$ means 3 clusters.

Step 2: Initialize Centroids Select K random points as centroids.

Step 3: Assign Points to Nearest Centroid Euclidean distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

For 3-dimensional data:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Step 4: Update Centroids New centroid:

$$C = \left(\frac{\sum x_i}{n}, \frac{\sum y_i}{n}, \frac{\sum z_i}{n} \right)$$

Step 5: Repeat Continue assigning and updating until clusters stop changing.

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Dataset for K-Means and PCA

ID	X	Y	Z
1	1	2	1
2	2	1	2
3	1	1	1
4	8	9	8
5	9	8	9
6	8	8	9
7	4	5	4
8	5	4	5
9	4	4	5
10	5	5	4

Question 1: Initial Clustering

Take:

- ▶ $K = 2$
- ▶ Initial centroids:
 - ▶ $C_1 = (1, 2, 1)$
 - ▶ $C_2 = (8, 9, 8)$

Tasks:

1. Compute Euclidean distance for each point.
2. Assign clusters.
3. Find updated centroids.

Question 2: Effect of Changing K

Using the same dataset:

1. Apply K-Means with $K = 2$.
2. Apply K-Means with $K = 3$.
3. Compare:
 - ▶ Number of clusters
 - ▶ Cluster compactness
 - ▶ Points changing clusters

Question 3: Final Cluster Identification

Suppose final centroids are:

$$C_1 = (1.33, 1.33, 1.33)$$

$$C_2 = (4.5, 4.5, 4.5)$$

$$C_3 = (8.33, 8.33, 8.67)$$

Tasks:

1. Identify cluster membership.
2. Explain why the grouping makes sense.

Question 4: Outlier Effect

Add the point:

(20, 20, 20)

Tasks:

1. Apply K-Means with $K = 3$.
2. Observe centroid movement.
3. Explain effect of outliers.

Question 5: Distance Computation

Find Euclidean distance between:

$$A = (1, 2, 1)$$

and

$$B = (5, 5, 4)$$

Solution setup:

$$d = \sqrt{(5 - 1)^2 + (5 - 2)^2 + (4 - 1)^2}$$

Advantages of K-Means

- ▶ Simple and fast
- ▶ Easy to implement
- ▶ Works well for spherical clusters
- ▶ Efficient for large datasets

Limitations of K-Means

- ▶ Need to choose K manually
- ▶ Sensitive to initialization
- ▶ Sensitive to outliers
- ▶ Poor performance on non-spherical clusters

Choosing Optimal K

One common method is the Elbow Method.

Idea:

- ▶ Compute Within Cluster Sum of Squares (WCSS)
- ▶ Plot WCSS vs K
- ▶ Choose elbow point

What is PCA?

PCA is a dimensionality reduction technique.

It transforms high-dimensional data into fewer dimensions while preserving maximum variance.

Example:

- ▶ Original features = 3
- ▶ PCA output = 2 principal components

Why PCA?

PCA helps in:

- ▶ Reducing computation
- ▶ Removing redundancy
- ▶ Visualization of high-dimensional data
- ▶ Noise reduction

Intuition Behind PCA

Suppose features are highly correlated.

Instead of using all features separately, PCA creates new variables called Principal Components.

These:

- ▶ Are uncorrelated
- ▶ Capture maximum variance

Step 1: Standardization

Standardization formula:

$$z = \frac{x - \mu}{\sigma}$$

where:

- ▶ μ = mean
- ▶ σ = standard deviation

Step 2: Covariance Matrix

Covariance formula:

$$\text{Cov}(X, Y) = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{n - 1}$$

Step 3: Eigenvalues and Eigenvectors

- ▶ Eigenvectors give principal directions.
- ▶ Eigenvalues represent importance of those directions.

Largest eigenvalue corresponds to Principal Component 1.

Step 4: Select Principal Components

Example:

- ▶ PC1 captures 85% variance
- ▶ PC2 captures 10% variance
- ▶ PC3 captures 5% variance

Then retain only PC1 and PC2.

Step 5: Transform Data

Project original data onto selected principal components.

Question 1: Mean Calculation

Compute:

- ▶ $\text{Mean}(X)$
- ▶ $\text{Mean}(Y)$
- ▶ $\text{Mean}(Z)$

Question 2: Standardization

For row 1:

$$(1, 2, 1)$$

Tasks:

1. Compute standardized values.
2. Explain importance of standardization.

Question 3: Covariance Interpretation

Suppose covariance matrix is:

	X	Y	Z
X	1	0.92	0.95
Y	0.92	1	0.90
Z	0.95	0.90	1

Tasks:

1. Identify strongly correlated features.
2. Explain why PCA is useful.

Question 4: Principal Component Selection

Suppose eigenvalues are:

Component	Eigenvalue
PC1	2.7
PC2	0.2
PC3	0.1

Tasks:

1. Which component captures maximum variance?
2. How many PCs should be retained?
3. Compute variance percentage captured by PC1.

Formula:

$$\text{Variance \%} = \frac{\text{Eigenvalue}}{\text{Total Eigenvalues}} \times 100$$

Question 5: Dimensionality Reduction

Original dataset:

- ▶ 3 dimensions

After PCA:

- ▶ Reduced to 2 dimensions

Tasks:

1. What information may be lost?
2. What are advantages of dimensionality reduction?

Question 6: PCA for Visualization

Explain:

1. Why PCA is useful for visualization.
2. How 100-dimensional data can be visualized in 2D.

Question 1

Can PCA be applied before K-Means?
Explain why.

Question 2

Why is feature scaling important in both K-Means and PCA?

Question 3

Suppose one feature has values from 1–10 and another from 1–10000.

Tasks:

1. Explain impact on K-Means.
2. Explain impact on PCA.
3. Why should normalization or standardization be used?

K-Means Summary

- ▶ Unsupervised clustering algorithm
- ▶ Groups similar data points
- ▶ Uses Euclidean distance
- ▶ Requires choosing K

PCA Summary

- ▶ Dimensionality reduction technique
- ▶ Finds principal components
- ▶ Preserves maximum variance
- ▶ Helps visualization and compression